

Jie-Ying (Jay) Lee

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RESEARCH INTERESTS

Moving from 3D vision and generative 3D (NeRF, 3D Gaussian Splatting) toward robot learning: (1) unified, flexible interfaces that let existing robot policies take in new sensing modalities such as vision, touch, and sound; (2) robots that continually improve themselves after deployment.

EDUCATION

- **National Yang Ming Chiao Tung University** Hsinchu, Taiwan
Ph.D. Student in Computer Science (Direct-Entry); withdrawing Sep. 2025 – Jan. 2027 (expected)
- **National Yang Ming Chiao Tung University** Hsinchu, Taiwan
B.S. in Computer Science Jul. 2021 – Jun. 2025
GPA: 4.23/4.3 (Overall), 4.26/4.3 (Major); Ranking: 10/184 (4.42%)
- **ETH Zurich** Zurich, Switzerland
Exchange Student, Department of Computer Science Sep. 2024 – Feb. 2025
Courses: Computer Graphics (6.0/6.0), Visual Computing (6.0/6.0), Computer Vision (5.5/6.0)

PUBLICATIONS

* denotes equal contribution; † denotes corresponding author.

- **Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery**
Jie-Ying Lee, Yi-Ruei Liu, Shr-Ruei Tsai, Wei-Cheng Chang, Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, Yu-Lun Liu
European Conference on Computer Vision (ECCV), 2026. [Project] [arXiv]
Synthesizes photorealistic, city-block scale 3D urban scenes from satellite imagery using diffusion models, eliminating the need for expensive 3D scanning and manual annotation while enabling real-time exploration.
- **YoCausal: How Far is Video Generation from World Model? A Causality Perspective**
You-Zhe Xie, Yu-Hsuan Li, Jie-Ying Lee, Kaipeng Zhang, Yu-Lun Liu, Zhixiang Wang
arXiv preprint, 2026. Under review. [Project] [arXiv]
- **Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion**
Ting-Hsuan Chen*, Ying-Huan Chen*, Tao Tu, Jie-Ying Lee, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo†, Yu-Lun Liu†, Yue Wang†, Liu Ren
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [Project] [arXiv]
- **BRDFusion: Physics Meets Generation for Urban Scene Inverse Rendering**
Yi-Ruei Liu, Jie-Ying Lee, Zheng-Hui Huang, Yu-Lun Liu†, Chih-Hao Lin†
arXiv preprint, 2026. Under review. [Project] [arXiv]
- **LightsOut: Diffusion-based Outpainting for Enhanced Lens Flare Removal**
Shr-Ruei Tsai, Wei-Cheng Chang, Jie-Ying Lee, Chih-Hai Su, Yu-Lun Liu
IEEE/CVF International Conference on Computer Vision (ICCV), 2025. [Project] [arXiv]
- **See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation**
Chih-Yao Hu*, Yang-Sen Lin*, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, Shr-Ruei Tsai, Chin-Yang Lin, Kuan-Wen Chen, Tsung-Wei Ke, Yu-Lun Liu
Conference on Robot Learning (CoRL), 2025. [Project] [arXiv]
- **AuraFusion360: Augmented Unseen Region Alignment for Reference-based 360° Unbounded Scene Inpainting**
Chung-Ho Wu*, Yang-Jung Chen*, Ying-Huan Chen, Jie-Ying Lee, Bo-Hsu Ke, Chun-Wei Tuan Mu, Yi-Chuan Huang, Chin-Yang Lin, Min-Hung Chen, Yen-Yu Lin, Yu-Lun Liu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [Project] [arXiv]
- **SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes**
Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu, Jie-Ying Lee, Jiun-Long Huang, Yu-Chee Tseng, Yu-Lun Liu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [Project] [arXiv]
- **BoostMVSNeRFs: Boosting MVS-based NeRFs to Generalizable View Synthesis in Large-scale Scenes**
Chih-Hai Su*, Chih-Yao Hu*, Shr-Ruei Tsai*, Jie-Ying Lee*, Chin-Yang Lin, Yu-Lun Liu
ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH), 2024. [Project] [arXiv]

EXPERIENCES

- **Google** – *Software Engineer, Pixel Camera Team* Sep. 2025 – Present
Building on-device perception for computational photography. Mentoring an intern on real-time camera composition control driven by LLM-generated state machines.
- **CompPhoto Lab, NYCU** – *Research Assistant and System Administrator* Aug. 2023 – Present
Advised by Prof. Yu-Lun Liu. Working on 3D scene synthesis, generative models for vision, and embodied AI, with a focus on NeRF, 3D Gaussian Splatting, and vision-language navigation. Led Skyfall-GS (ECCV 2026), which synthesizes immersive 3D urban scenes from satellite imagery. Currently working on interfacing pretrained robot policies with new sensor modalities.
- **Google** – *Software Engineering Intern, Pixel Camera Team* Jun. 2024 – Sep. 2024
Brought the Segment Anything Model (SAM) to mobile devices by distilling it into a lightweight MobileNetV4, with only a 4% performance drop and 20 ms latency (8x faster) on a mobile TPU.
- **IT Center of CS Department, NYCU** – *Teaching Assistant, WWW and System Group* Jul. 2022 – Jun. 2025
Built the backend of the department's TA contact information system, including database schema design, RESTful APIs, and API documentation, following a merge-request and CI/CD workflow.
- **Microsoft** – *Research & Development Intern, Bing Geocoding* Sep. 2023 – Jun. 2024
Extended a backend component to support a new data format with dashboard visualization; validated the DeepCAL address-parsing model and improved its training pipeline.
- **Appier** – *Backend Engineer Summer Intern* Jul. 2023 – Sep. 2023
Built a system that automatically computes and visualizes per-customer, per-feature cloud costs, replacing manual estimation and achieving a 7200x speedup; gathered requirements from PMs and sales.

ACADEMIC SERVICE

- **Reviewer:** ECCV 2026, SIGGRAPH Asia 2026, ACCV 2026, WACV 2027, IJCV
- **Teaching Assistant:** Image and Video Generation, NYCU (2025 Fall)

AWARDS

- **Shun-I Chu Zyxel Scholarship:** Awarded to one of only two computer science students for exceptional academic performance (2024).
- **NSTC Research Grant for University Students:** Awarded the undergraduate research grant by the National Science and Technology Council, Taiwan (2024).
- **Third Place of the NYCU CS Undergraduate Research Competition:** Awarded for the project "BEVGaussian: Generate Scene-level 3D Gaussian from BEV image" (2024).
- **Bronze Award of The 2022 ICPC Asia Taoyuan Regional Programming Contest:** Ranked 32nd of 100 teams; second among bronze-award teams.
- **Academic Achievements Award in 2021 Fall Semester:** Top 5% in the class.
- **Fundamental Course Award of Discrete Mathematics, Digital Circuit Design, Data Structures & Object-oriented Programming and Introduction to Algorithms:** Top 5% of these courses.
- **Macronix Award of 19th Macronix Science Award:** Top award in Macronix Science Award. The only recipient within ten years.
- **Second Place in 60th National Primary and High School Science Fair:** Awarded the best work at the 60th national science fair (The first place is absent) out of 8 all participants who are the first place from cities in Taiwan.
- **Fourth Prize in 2021 Taiwan International Science Fair:** Awarded fourth prize out of 21 international participants.

PROJECTS

- **Physically-based Renderer (ETH Zurich Computer Graphics):** Extended the Nori ray tracer in C++ with Monte Carlo path tracing, multiple importance sampling, and a dielectric BSDF. In a two-person final project, implemented heterogeneous participating media with single- and multiple-scattering volumetric path tracers to render a waterfall cave scene; selected for the final round of the course rendering competition. (Dec. 2024) [\[Blog\]](#)
- **Operating System Capstone: Bare-metal Kernel on Raspberry Pi:** Implemented a bare-metal OS kernel in C and assembly for Raspberry Pi 3B+ (ARMv8-A, AArch64) across eight labs: UART shell and mailbox interface, a self-hosted UART bootloader with device tree parsing, exception and interrupt handling, a buddy-system memory allocator, threads and user processes with system calls and signals, virtual memory, and a virtual file system with tmpfs, initramfs, and device file systems. (Jun. 2024) [\[Code\]](#)
- **P Language Compiler (NYCU Compiler Design):** Built a complete compiler in C++ for P, a Pascal-like language, using flex and bison: scanner, parser, AST construction, semantic analysis with symbol tables, and RISC-V code generation executed on the Spike simulator. (Jan. 2024) [\[Code\]](#)